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Roll No

AU-7003-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Computer Aided Engineering

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Assume suitable missing data if any.

1. a) What is CAD? Write short notes on input devices, output devices and graphics standards.
b) Enlist various standard functioning at various level of graphic system. Explain IGES in brief.
2. a) Explain nodal shape functions of 1-D bar.
b) What are synthetic curves? Explain need of parametric representation of synthetic curves.
3. Find a parametric cubic curve that starts at $P_0(-1, 2)$, ends at $P_3(8, 4)$ and passes through two prescribed points $P_1(2, 4)$ and $P_2(6, 6)$.
4. a) Derive Bezier polynomial equation for curves considering Bernstein polynomials as basis function.
b) Enlist various synthetic curves used for surface modeling. Explain any one with mathematical representations.

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5. a) Explain boundary representation in solid modelling.
b) Explain importance of Euler's equation in solid modelling.
6. a) Solve the given simultaneous equation using Gaussian elimination approach.

$$x_1 - 2x_2 + 6x_3 = 0$$

$$2x_1 + 2x_2 + 3x_3 = 3$$

$$-x_1 + 3x_2 = 3$$

- b) Enlist the step involved in solution of a general continuum problem by finite element method.
7. a) With neat sketch explain surface of revolution and tabulated type analytical surfaces.
b) Show mathematically that a bicubic surface patch degenerates to a cubic spline if the four corners of the patch are collapsed to two.
8. Write short notes on any two of the following:
 - a) Graphic standards
 - b) Visual realism
 - c) Sweep representation
 - d) Aspect ratio
 - e) Stiffness matrix for one dimensional analysis
